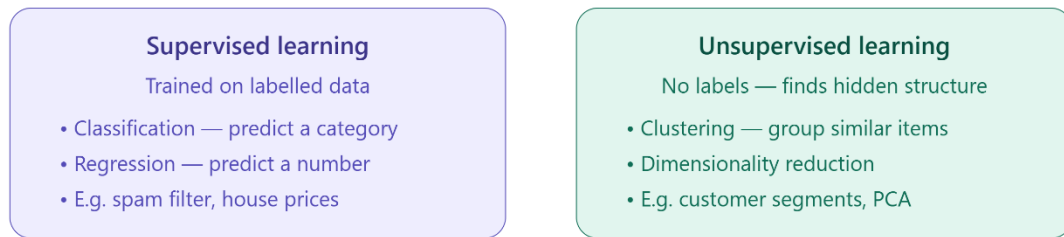
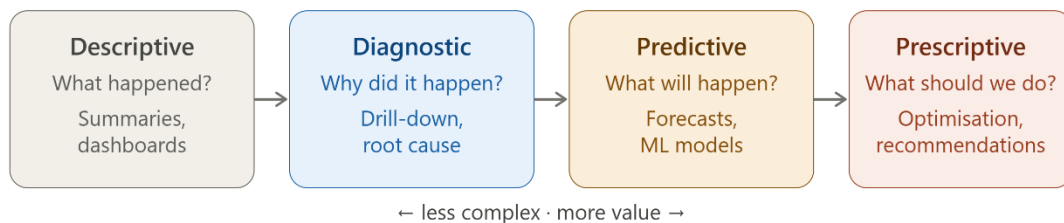


ML Fundamentals: Supervised vs Unsupervised, Types of Analytics, Regression Fundamental

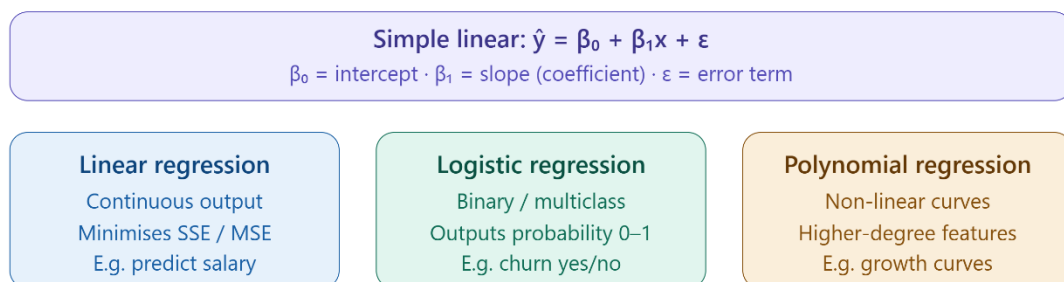
Machine learning: two core paradigms



Four types of analytics



Regression fundamentals



Here's a breakdown of all three topics:

Supervised vs Unsupervised Learning

The key distinction is whether your training data has labels. In supervised learning, every example has a known answer — the model learns to map inputs to outputs, which is the basis for both classification (predicting a class like "spam" or "not spam") and regression (predicting a value like house price). In unsupervised learning, there are no labels; the model discovers structure on its own, most commonly by grouping similar data points (clustering) or compressing many features into fewer (dimensionality reduction).

A third paradigm worth knowing is **reinforcement learning**, where an agent learns by trial and error via rewards — used in game AI and robotics.

Four Types of Analytics

These represent increasing analytical sophistication and business value:

- **Descriptive** — what your data says about the past. Averages, totals, bar charts. The foundation everything else builds on.

- **Diagnostic** — *why* something happened. Involves slicing data, correlation analysis, and identifying root causes.
- **Predictive** — using statistical models or ML to forecast what's likely to happen. This is where regression and classification models live.
- **Prescriptive** — recommending an action to optimize an outcome. Think recommendation engines or supply chain optimization solvers.

Regression Fundamentals

Regression models a relationship between input features (X) and a continuous output (y). Three key types:

- **Linear regression** fits a straight line through data, minimizing the sum of squared errors (SSE). Interpretable and fast — a good first model to try.
- **Logistic regression** (despite the name) is a classifier — it applies a sigmoid function to output a probability between 0 and 1, then thresholds it to predict a class.
- **Polynomial regression** extends linear regression by adding squared or cubed features, capturing curves in the data — but watch for overfitting.

Key metrics to evaluate regression models: **R^2** (how much variance the model explains), **MSE/RMSE** (average squared/root-squared error), and **MAE** (average absolute error).